NF White Delay

PROFESSIONAL DELAY

Version 1.0.0 · NF Audio Tools — By Nenno Fernando · VST3 / AU / AAX

1. Overview

NF White Delay is a tempo-synced stereo delay built around a clean digital core with two optional colour engines — **Analog** and **Tape** — plus modulation, filtering and a musical ducking/character stage. It is designed to sit comfortably on a vocal, a guitar bus or a full mix send: transparent when you need it clean, characterful when you push the Character and Mode controls.

Its signal path is: **Input** → **Delay line (Digital / Analog / Tape mode, Ping-Pong optional)** → **Modulation (chorus-style pitch wobble on the repeats)** → **Filters (High Pass / Low Pass / Resonance shaping the repeats)** → **Character & Ducking** → **Dry/Wet** → **Output**.

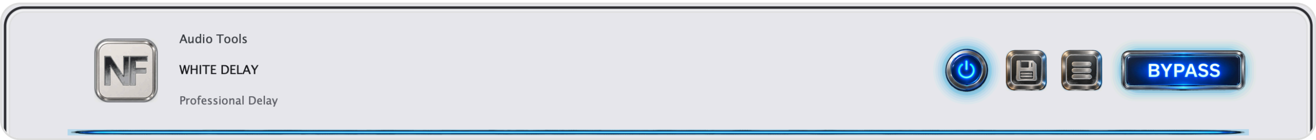

Fig. 1 — Header: brand, Save, Menu (opens this manual), Bypass.

2. Main Section

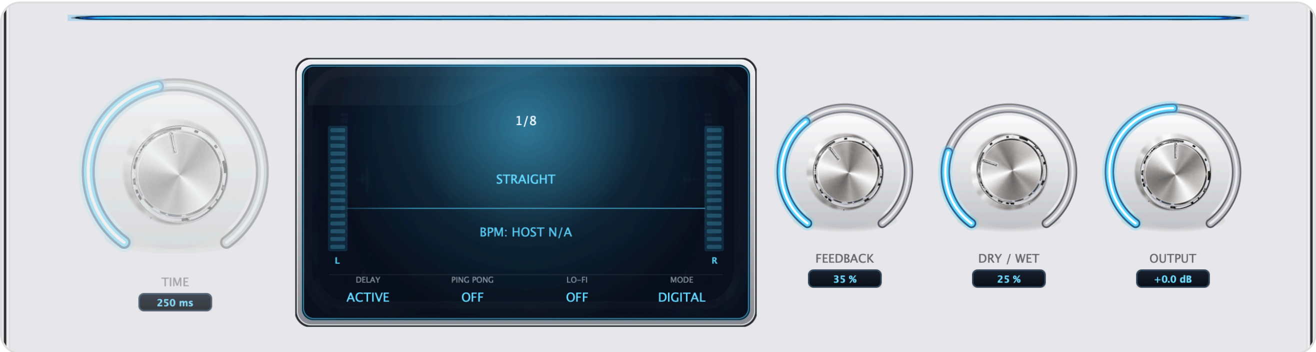

Fig. 2 — TIME, live display, FEEDBACK, DRY/WET and OUTPUT.

Control	Description
TIME	Delay time in milliseconds (1–2000 ms). Dimmed and read-only when SYNC is on, since the division/modifier below take over.
Display	Shows the active division (or free-running ms), SYNC modifier (Straight/Dotted/Triplet), host BPM, and a live stereo activity meter of the wet signal (L/R).
FEEDBACK	How much of each repeat feeds back into the delay line -- higher values create longer, denser repeat trails.
DRY / WET	Balance between the untouched input and the delayed signal.
OUTPUT	Final output trim in dB, applied after the Dry/Wet mix.

3. Sync & Quick Controls

Fig. 3 — SYNC, DIVISION, MODIFIER, PING PONG / LO-FI, DIGITAL / ANALOG / TAPE.

Control	Description
SYNC	Locks the delay time to the host tempo using DIVISION and MODIFIER instead of the free TIME knob.
DIVISION	Note value used when SYNC is on: 1/64 up to 2 Bars.
MODIFIER	Straight, Dotted or Triplet feel applied to the selected division.
PING PONG	Alternates repeats between the left and right channels instead of a centred delay.
LO-FI	Adds bit-crush/sample-rate style degradation to the repeats for a lower-fidelity, vintage character.
DIGITAL / ANALOG / TAPE	Delay engine character: Digital is the clean reference; Analog adds warmth and gentle non-linearity typical of BBD-style delays; Tape adds wow/flutter and saturation typical of tape echo units.

4. Modulation, Filters & Character

Fig. 4 — The three lower modules: Modulation, Filters, Character.

Modulation

Control	Description
RATE	Speed of the pitch-modulation LFO applied to the repeats, in Hz.
DEPTH	Amount of pitch modulation -- higher settings create a stronger chorus/vibrato effect on the repeats.
SHAPE	LFO waveform: Sine, Triangle, or Soft Random for a less predictable, tape-like wobble.
SPREAD	Stereo offset between the left and right modulation phase, widening the repeats.

Filters

Control	Description
HIGH PASS	Removes low frequencies from the repeats, keeping the low end of the dry signal clean.
LOW PASS	Removes high frequencies from the repeats, for a darker, more distant trail.
RESO	Resonance/emphasis at the filter cutoffs, for a more pronounced, characterful roll-off.

Character

Control	Description
AMOUNT	Overall intensity of the Analog/Tape colour stage (saturation, wobble, age) on the repeats.
DUCKING	Sidechains the repeats against the dry input, so the delay ducks under the source and comes forward in the gaps -- keeps busy passages readable.

> Tip: for a classic slap-back, turn SYNC off, set TIME between 80–140 ms, keep FEEDBACK low, and use DIGITAL mode with Character at zero.

5. Header & Bypass

- **Save icon** -- preset management (coming in a future update).
- **Menu icon (three lines)** -- opens this manual and the About box.
- **BYPASS** / power button -- both toggle the same true bypass; the delay line is frozen and the dry signal passes through untouched.